Práctica 3.2 Wazuh con DHCP

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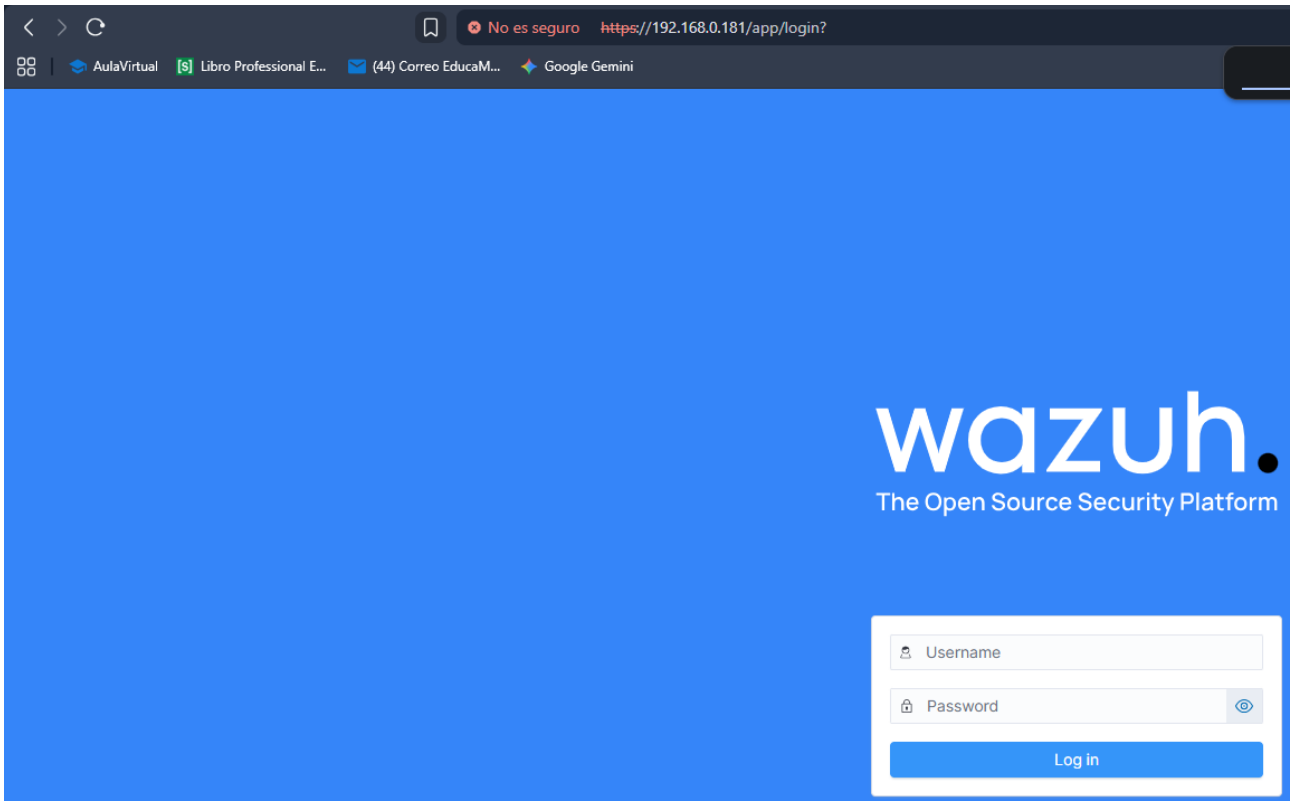
1.1 Posibles errores

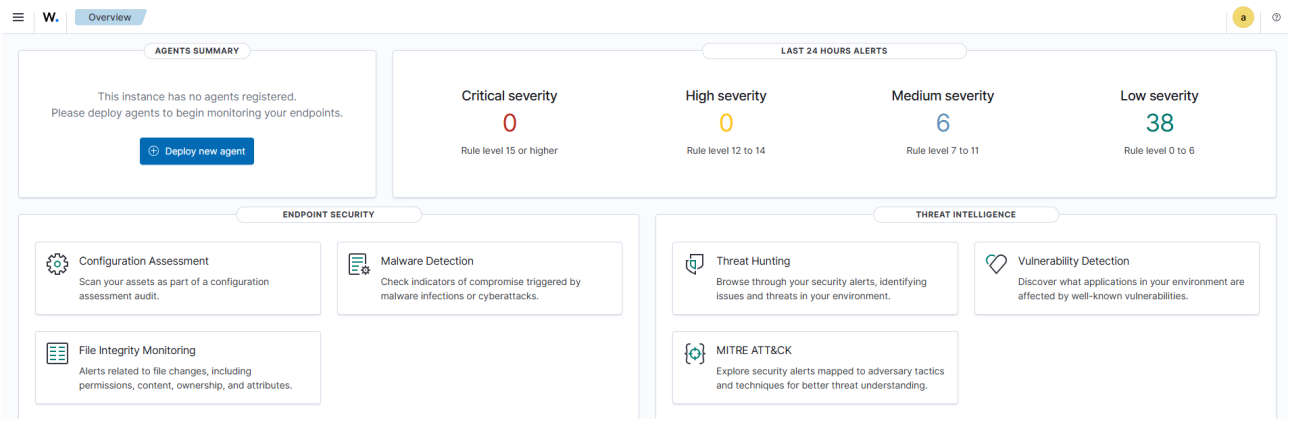
He importado la OVA de Wazuh y como puede dar error por bloqueo, me conecto por SSH desde cmd:

```
C:\Users\ASIR2> ssh  wazuh-user@192.168.0.181
The authenticity of host '192.168.0.181 (192.168.0.181)' can't be established.
ECDSA key fingerprint is SHA256:7s4ojPDCQjoHJbusnB5nI5dE58bkLK/H63ysnpvhD2U.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.0.181' (ECDSA) to the list of known hosts.
wazuh-user@192.168.0.181's password:
wwwwww.            wwwwwww.           wwwwwww.
wwwwwww.           wwwwwww.           wwwwwww.
 wwwwww.          wwwwwwwww.         wwwwwww.
 wwwwwww.         wwwwwwwww.         wwwwwww.
  wwwwww.        wwwwwwwwwww.       wwwwwww.
  wwwwwww.       wwwwwwwwwww.       wwwwwww.
   wwwwww.      wwwwww.wwwwww.     wwwwwww.
   wwwwwww.     wwwww. wwwwww.     wwwwwww.
    wwwwww.    wwwwww.  wwwwww.   wwwwwww.
    wwwwwww.   wwwww.   wwwwww.   wwwwwww.
     wwwwww.  wwwwww.     wwwwww.wwwwwww.
     wwwwwww.wwwww.      wwwwww.wwwwwww.
      wwwwwwwwwwwww.       wwwwwwwwwwwww.
      wwwwwwwwwwww.        wwwwwwwwwwwww.       ooooo
       wwwwwwwwwww.         wwwwwwwwwww.       ooooooo
       wwwwwwwwww.          wwwwwwwwwww.      ooooooooo
        wwwwwwwww.           wwwwwwwww.       ooooooooo
        wwwwwwww.            wwwwwwwww.        ooooooo
         wwwwwww.             wwwwwww.          ooooo
```

1.2 Conectarse a Wazuh a través del navegador



La dirección 192.168.0.181 es de la máquina Wazuh (admin/admin)



1.3 Instalación de los agentes

```
root@debian:/home/usuario# sudo apt install curl
```

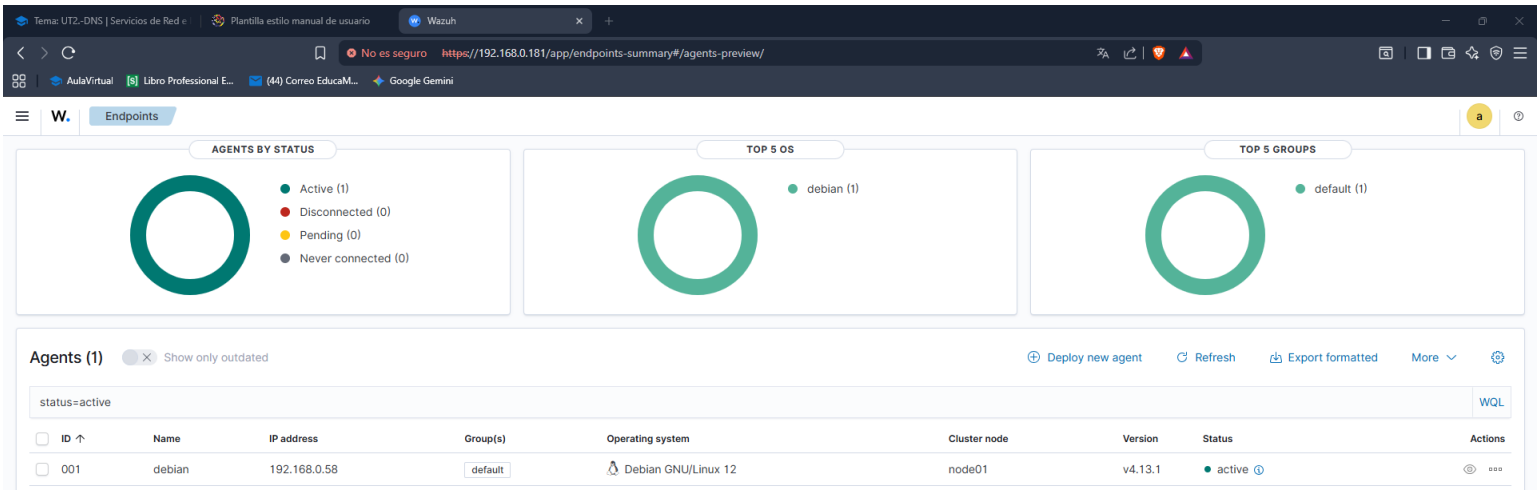
```
root@debian:/home/usuario# curl -s https://packages.wazuh.com/key/GPG-KEY-WAZUH
| gpg --no-default-keyring --keyring gnupg-ring:/usr/share/keyrings/wazuh.gpg --
import && chmod 644 /usr/share/keyrings/wazuh.gpg
```

```
deb [signed-by=/usr/share/keyrings/wazuh.gpg] https://packages.wazuh.com/4.x/apt/ stable main
```

```
root@debian:/home/usuario# WAZUH_MANAGER="192.168.0.181" apt-get install wazuh-agent
```

La IP 192.168.0.181 es la de mi máquina Wazuh

```
root@debian:/etc/apt# systemctl daemon-reload
root@debian:/etc/apt# systemctl enable wazuh-agent
Created symlink /etc/systemd/system/multi-user.target.wants/wazuh-agent.service → /lib/system
d/system/wazuh-agent.service.
root@debian:/etc/apt# systemctl start wazuh-agent
```



Sale la dirección 192.168.0.58 que es la de mi máquina agente y sale activa por lo que el servicio de Wazuh funciona correctamente.

2. Atacar al agente

En la maquina Kali Linux usamos Hydra para atacar a la maquina agente:

Me intento conectar al agente a través de SSH (he instalado en el agente openssh-server para conectarme por SSH):

```
┌──(root㉿osboxes)-[~]
└─# ssh usuario@192.168.0.58
The authenticity of host '192.168.0.58 (192.168.0.58)' can't be established.
ED25519 key fingerprint is SHA256:OmJoOVbmkt7CkCzbUKNm0l1NhSwKnns1SXCZM2or68w
.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.0.58' (ED25519) to the list of known host
s.
usuario@192.168.0.58's password:
Linux debian 6.1.0-40-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.153-1 (2025-09-
20) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
usuario@debian:~$
```

Funciona la conexión por SSH y he podido conectarme.

Voy a atacar con Hydra he creado dos .txt uno con usuarios y otro con contraseñas.

```
root@osboxes: ~
File  Actions  Edit  View  Help
  GNU nano 6.3                    usuarios.txt *
admin
host
local
wazuh
wazuh-user
```

```
root@osboxes: ~
File  Actions  Edit  View  Help
  GNU nano 6.3                    passwords.txt
123456
admin
wazuh
wazuh-server
root
```

Uso Hydra falla porque la contraseña y el usuario no estaban en los diccionarios:

```
┌──(root㉿osboxes)-[~]
└─# hydra -L usuarios.txt -P passwords.txt ssh://192.168.0.58
Hydra v9.3 (c) 2022 by van Hauser/THC & David Maciejak - Please do not use in
 military or secret service organizations, or for illegal purposes (this is n
on-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-10-24 06:
21:48
[WARNING] Many SSH configurations limit the number of parallel tasks, it is r
ecommended to reduce the tasks: use -t 4
[DATA] max 16 tasks per 1 server, overall 16 tasks, 25 login tries (l:5/p:0),
 ~5 tries per task
[DATA] attacking ssh://192.168.0.58:22/
1 of 1 target completed, 0 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2025-10-24 06:
21:50
```

Miro Wazuh y veo las conexiones que se han intentado por SSH, que son las que ha hecho Hydra:

timestamp	agent.name	rule.description	rule.level	rule.id
Oct 24, 2025 @ 12:29:49.3...	debian	sshd: Attempt to login using a non-existent user	5	5710
Oct 24, 2025 @ 12:29:49.3...	debian	sshd: Attempt to login using a non-existent user	5	5710
Oct 24, 2025 @ 12:29:49.3...	debian	sshd: Attempt to login using a non-existent user	5	5710
Oct 24, 2025 @ 12:29:49.3...	debian	sshd: Attempt to login using a non-existent user	5	5710
Oct 24, 2025 @ 12:29:47.4...	debian	sshd: Attempt to login using a non-existent user	5	5710
Oct 24, 2025 @ 12:29:47.4...	debian	sshd: Attempt to login using a non-existent user	5	5710
Oct 24, 2025 @ 12:29:47.4...	debian	sshd: Attempt to login using a non-existent user	5	5710
Oct 24, 2025 @ 12:29:47.4...	debian	sshd: authentication failed.	5	5760
Oct 24, 2025 @ 12:29:47.4...	debian	sshd: authentication failed.	5	5760
Oct 24, 2025 @ 12:29:47.4...	debian	sshd: authentication failed.	5	5760
Oct 24, 2025 @ 12:29:47.4...	debian	sshd: authentication failed.	5	5760
Oct 24, 2025 @ 12:29:47.3...	debian	sshd: Attempt to login using a non-existent user	5	5710
Oct 24, 2025 @ 12:29:47.3...	debian	sshd: Attempt to login using a non-existent user	5	5710
Oct 24, 2025 @ 12:29:47.3...	debian	sshd: Attempt to login using a non-existent user	5	5710
Oct 24, 2025 @ 12:29:47.3...	debian	sshd: Attempt to login using a non-existent user	5	5710

Ahora lo he intentado con el usuario y contraseña correctos en los diccionarios e hydra funciona correctamente (además en Wazuh me sale el evento de logeo existoso):

```
└─# sudo hydra -t 4 -L usuarios.txt -P passwords.txt ssh://192.168.0.58
Hydra v9.3 (c) 2022 by van Hauser/THC & David Maciejak - Please do not use in
 military or secret service organizations, or for illegal purposes (this is n
on-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-11-07 04:
42:34
[DATA] max 4 tasks per 1 server, overall 4 tasks, 36 login tries (l:6/p:0), ~
6 tries per task
[DATA] attacking ssh://192.168.0.58:22/
[22][ssh] host: 192.168.0.58   login: usuario   password: 123456
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2025-11-07 04:
42:41
```

Nov 7, 2025 @ 10:46:28.612	debian	Multiple authentication failures followed by a success.	12	40112
Nov 7, 2025 @ 10:46:28.610	debian	PAM: User login failed.	5	5503
Nov 7, 2025 @ 10:46:28.598	debian	sshd: Attempt to login using a non-existent user	5	5710

2.1 Respondiendo a Hydra

Para evitar el acceso repetido por SSH, en var/ossec/etc/ossec.conf del agente añado lo siguiente:

```
<command>
  <name>firewall-drop</name>
  <executable>firewall-drop</executable>
  <timeout_allowed>yes</timeout_allowed>
</command>
```

```
<ossec_config>
  <active-response>
    <disabled>no</disabled>
    <command>firewall-drop</command>
    <location>local</location>
    <rules_id>5763</rules_id>
    <timeout>180</timeout>
  </active-response>
</ossec_config>
```

Lo configurado hace que salte el evento 5763 que es el de Acceso por SSH por acceso a fuerza bruta y que además bloquea el acceso por SSH por 180 segundos.

Hago ping para ver que funciona correctamente y luego reinicio el servicio para que la conexión sea bloqueada:

```
┌──(root㉿osboxes)-[~]
└─# ping 192.168.0.58
PING 192.168.0.58 (192.168.0.58) 56(84) bytes of data.
64 bytes from 192.168.0.58: icmp_seq=1 ttl=64 time=0.627 ms
64 bytes from 192.168.0.58: icmp_seq=2 ttl=64 time=0.626 ms
64 bytes from 192.168.0.58: icmp_seq=3 ttl=64 time=0.678 ms
^C
--- 192.168.0.58 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2004ms
```

Reinicio el agente:

```
root@debian:/var/ossec/bin# sudo systemctl restart wazuh-agent
root@debian:/var/ossec/bin# sudo systemctl status wazuh-agent
● wazuh-agent.service - Wazuh agent
     Loaded: loaded (/lib/systemd/system/wazuh-agent.service; enabled; preset: enabled)
     Active: active (running) since Tue 2025-11-04 10:22:43 CET; 28s ago
```

Ataco con Hydra:

```
┌──(root㉿osboxes)-[~]
└─# sudo hydra -t 4 -L usuarios.txt -P passwords.txt 192.168.0.58 ssh
```

Miro en Wazuh y efectivamente me sale el evento:

Nov 4, 2025 @ 10:23:37.590	debian	sshd: brute force trying to get access to the system. Authentication failed.	10	5763

sshd: brute force trying to get access to the system. Authentication failed.	10	5763

Ahora si intento hacer ping a mi agente debería descartar los paquetes y efectivamente lo hace:

```
┌──(root㉿osboxes)-[~]
└─# ping 192.168.0.58
PING 192.168.0.58 (192.168.0.58) 56(84) bytes of data.
^C
--- 192.168.0.58 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3062ms
```